

Warm-Up: Solving Systems and Determinants

Name: _____

Date: _____

Part 1: Solve each system

Find all solutions, or explain why there are none.

1.

$$x + y = 10$$

$$x - y = 2$$

2.

$$x + y = 10$$

$$x + y = 12$$

3.

$$x + 2y + 3z = 8$$

$$2x - 3y - z = 10$$

$$2x + 4y + 6z = 9$$

Part 2: Matrix equations and determinants

- (a) Write each system from Part 1 as a matrix equation $A\mathbf{x} = \mathbf{b}$.
(b) Find the determinant of each coefficient matrix A .

1. Matrix equation:

$$\det(A) =$$

2. Matrix equation:

$$\det(A) =$$

3. Matrix equation:

$$\det(A) =$$

Part 3: Look back

What do your answers to Part 1 and your determinants in Part 2 have in common? Which determinants were zero, and what happened to the corresponding systems?